Overview Results

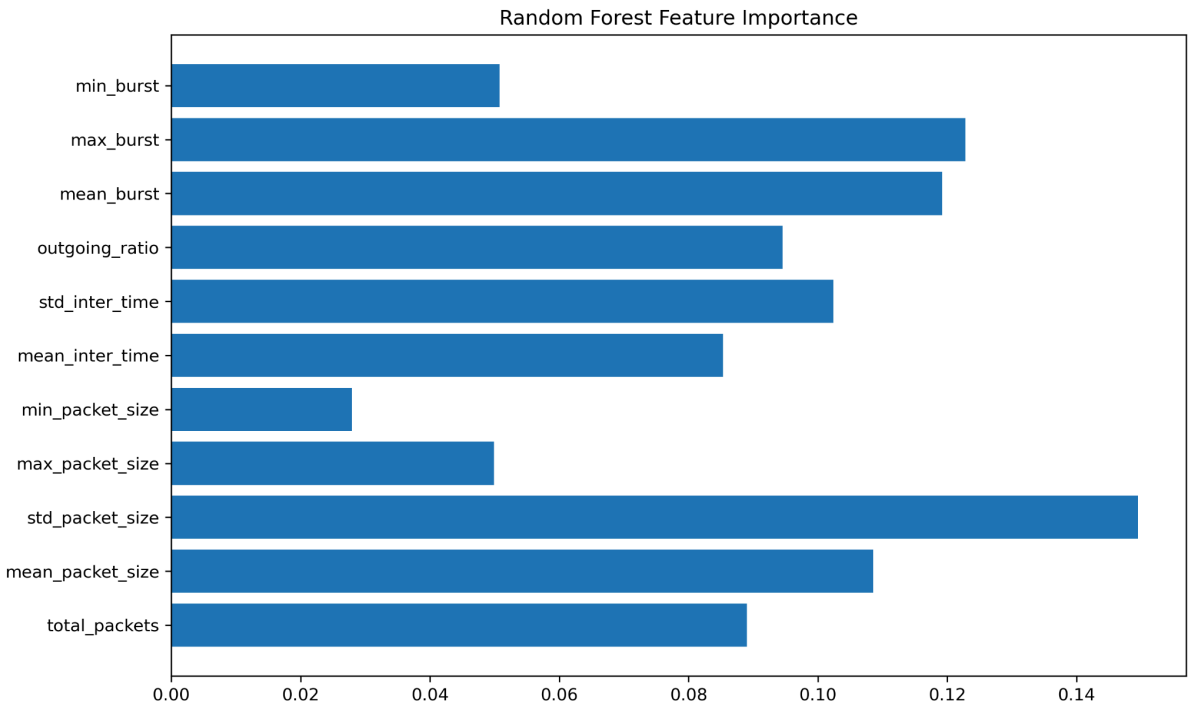
		Accuracy	Precision	Recall	F1-score
Model	**Datatype**				
RF	CSV	0.6375	0.5208	0.6375	0.5583
	NPZ	0.7111	0.7303	0.7111	0.7018
SVM	CSV	0.5875	0.4875	0.5875	0.5162
	NPZ	0.7668	0.7974	0.7668	0.7716
CNN	CSV	0.3125	0.3489	0.5	0.3896
CB	CSV	0.6375	0.5396	0.6375	0.5708
CNN	NPZ	0.6921	0.702	0.6921	0.6819

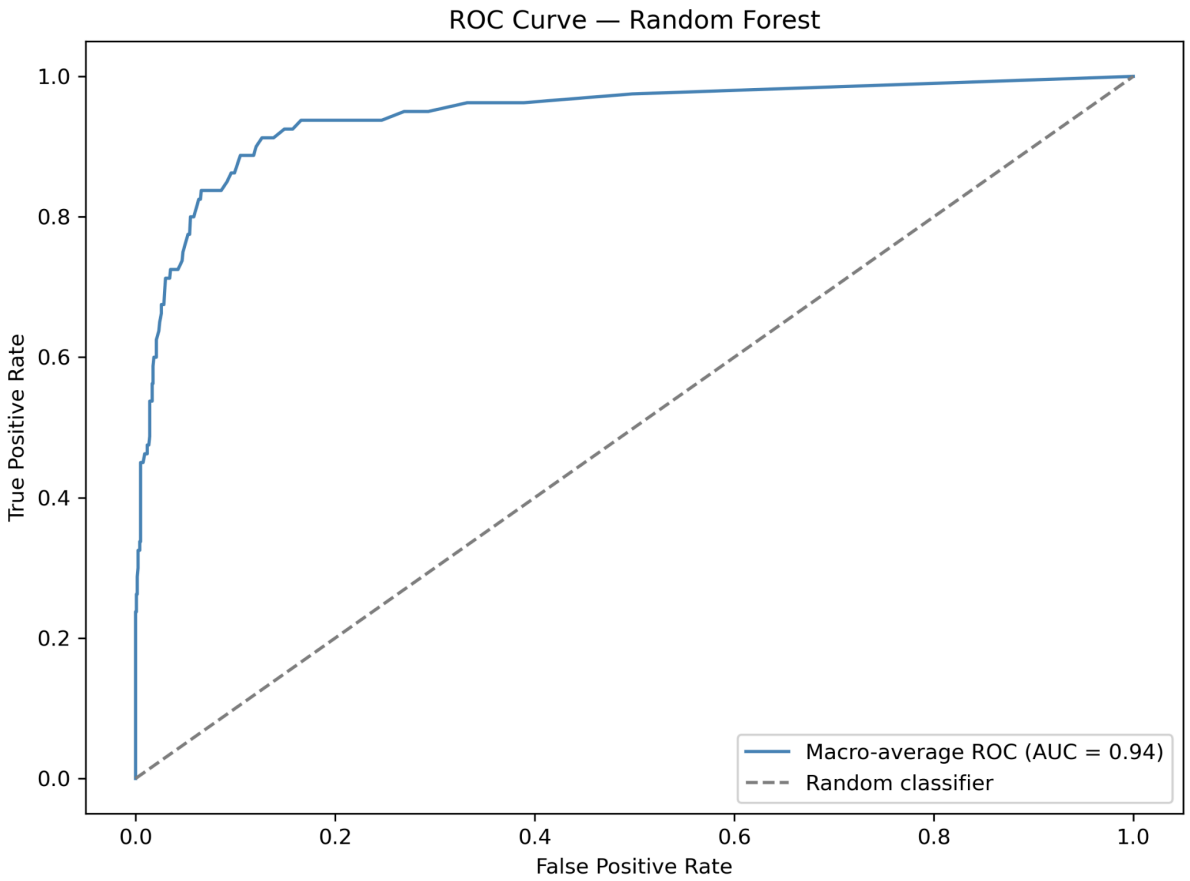
CSV dataset:

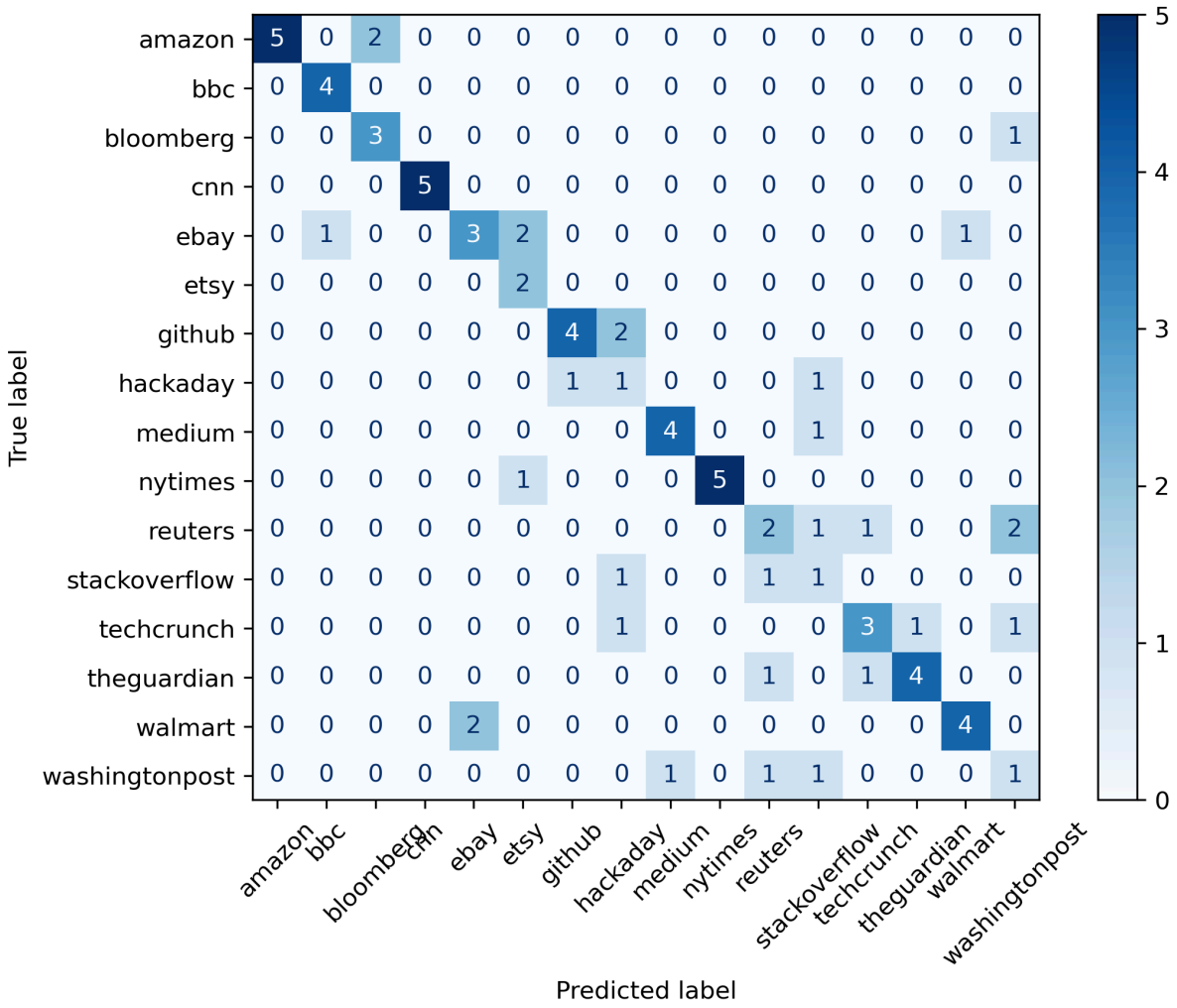
Random Forest:

Random Forest is perfectly suited for datasets with tabular statistical features like size of packets, time intervals, bursts, direction of packets. It works well with numerical and categorical data. Random Forest works better than Decision Tree as it handles small anf large feature sets well, it is resistant to noise and outliers which are present in case of Tor traffic, and scalable. Its main advantage is that it generates a random subset of features.
To ensure higher accuracy, I used cross-validation with 5 folds. This method guarantees less dependence on random training/testing separation and gives a more stable final result.

Feature Importance graph for RF shows that only a subset of features is rather important in traffic classification for Tor website. Those features are the range of packet sizes, max size of burst and mean size of burst. Those features distinguish different categories of websites. The diagonal in confusion matrix is clearly expressed, but we can see that there are misses between similar websites.
Among all trained models RF showed the most precise result on a small dataset because it is suited for statistical dependencies and it does not require feature extension and high data volumes.

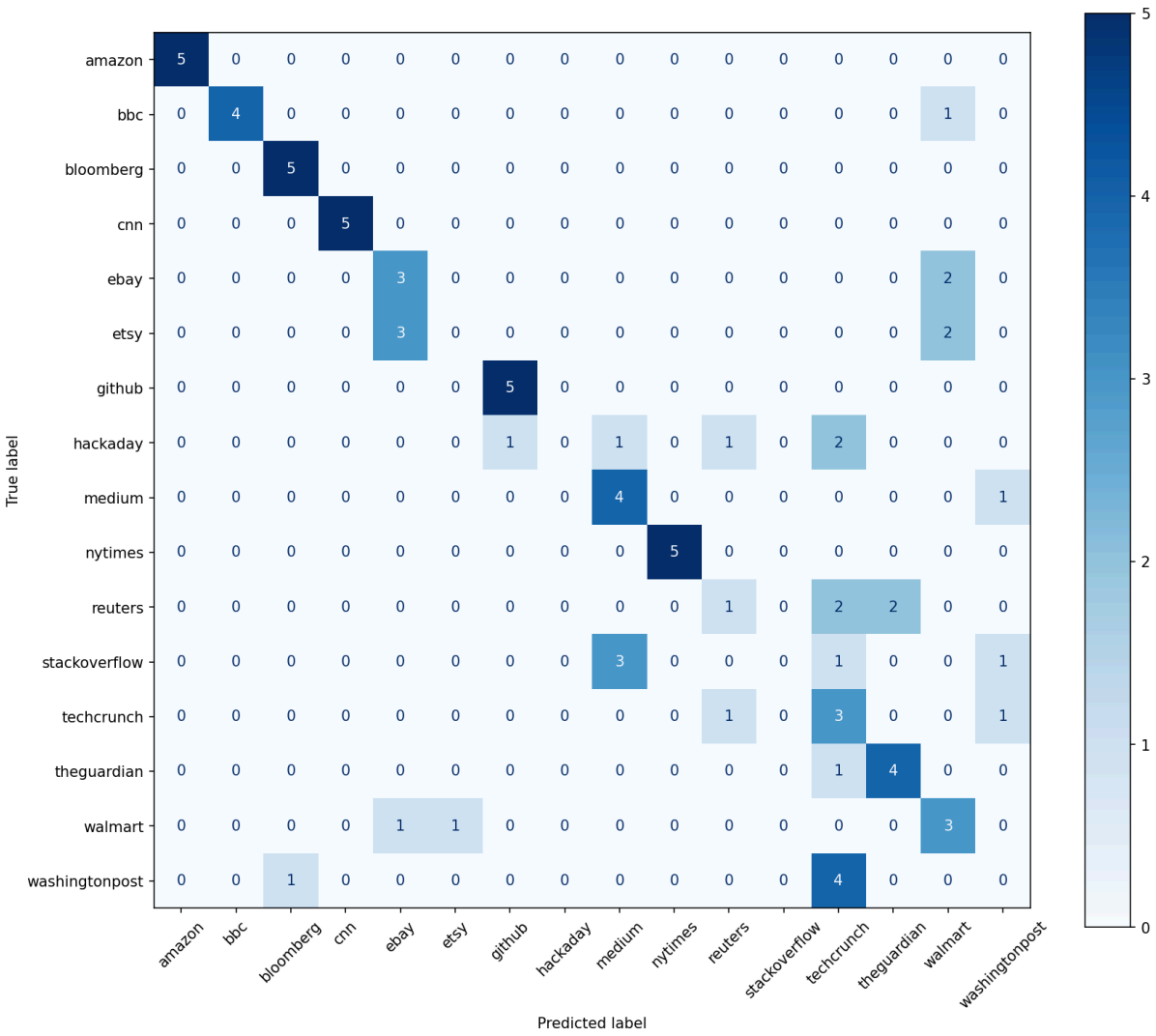

Random Forest Feature Importance
min_burst
max_burst
mean_burst
outgoing_ratio
std_inter_time
mean_inter_time
min_packet_size
max_packet_size
std_packet_size
mean_packet_size
total_packets
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14

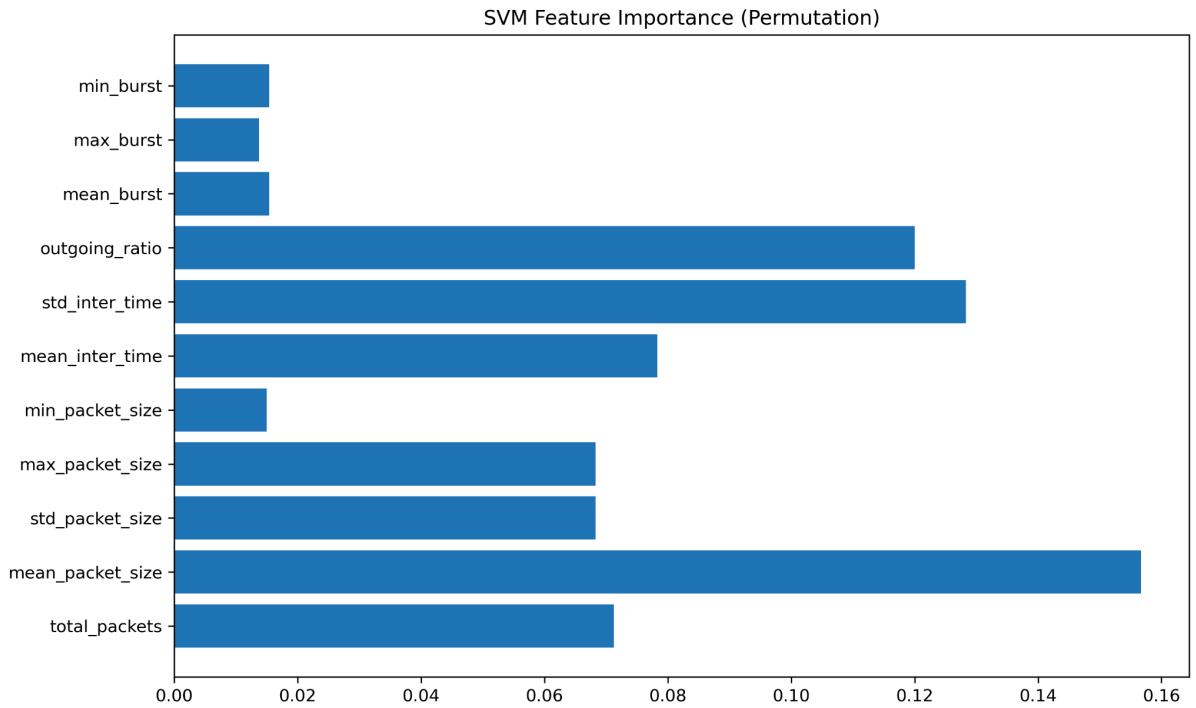

ROC Curve — Random Forest
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Macro-average ROC (AUC = 0.94)
Random classifier

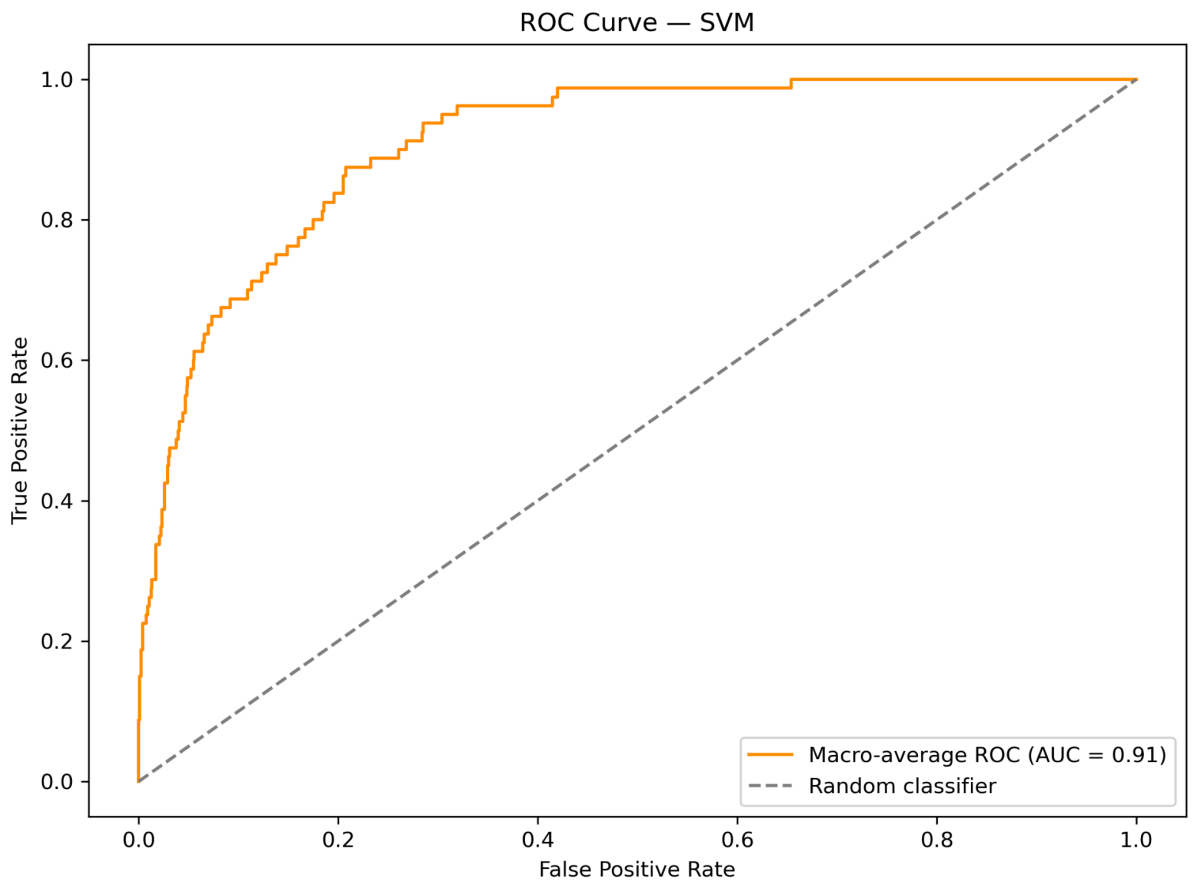


SVM

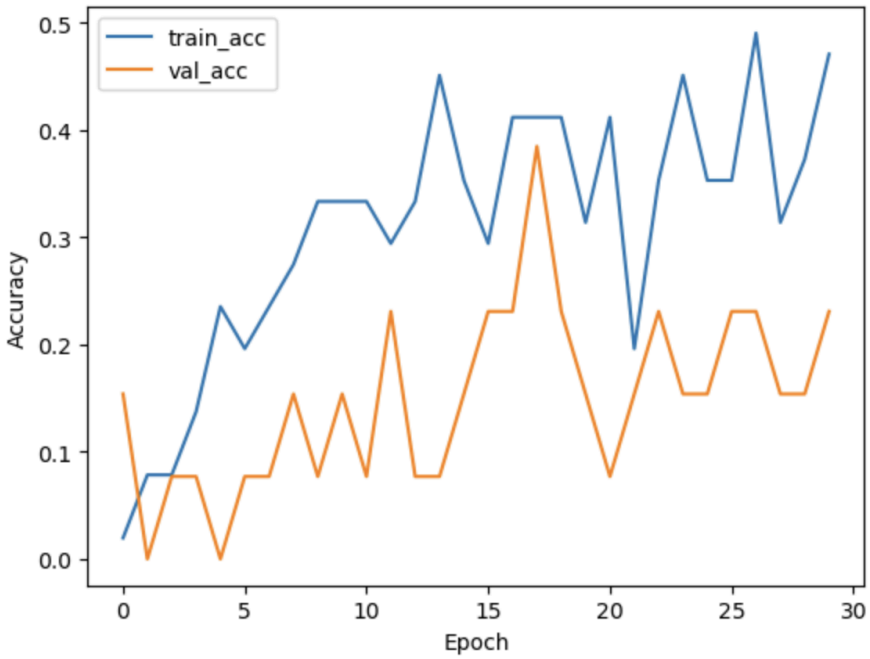
Support Vector Machine finds the best boundary that separates data points. In this research I used the RBF kernel which works with non-linear dependencies, like traffic patterns. SVM operates on high-dimensional data: time constraints, direction of packets, and size of packets. RBF kernel transforms non-linear into linear features to analyze with vectors. This model succeeded in distinguishing certain patterns and partially distributing websites according to their characteristics. The reason precision is lower than recall could be the presence of false positives, which means some websites are similar in traffic patterns. The confusion matrix diagonal is rather straight - it implies that the model trains and is able to separate different traffic categories. However, it is more sensitive to noise and outliers than Random Forest. For the feature importance, mean packet size, range in intervals' time, and outgoing packets ratio more more valuable to the model

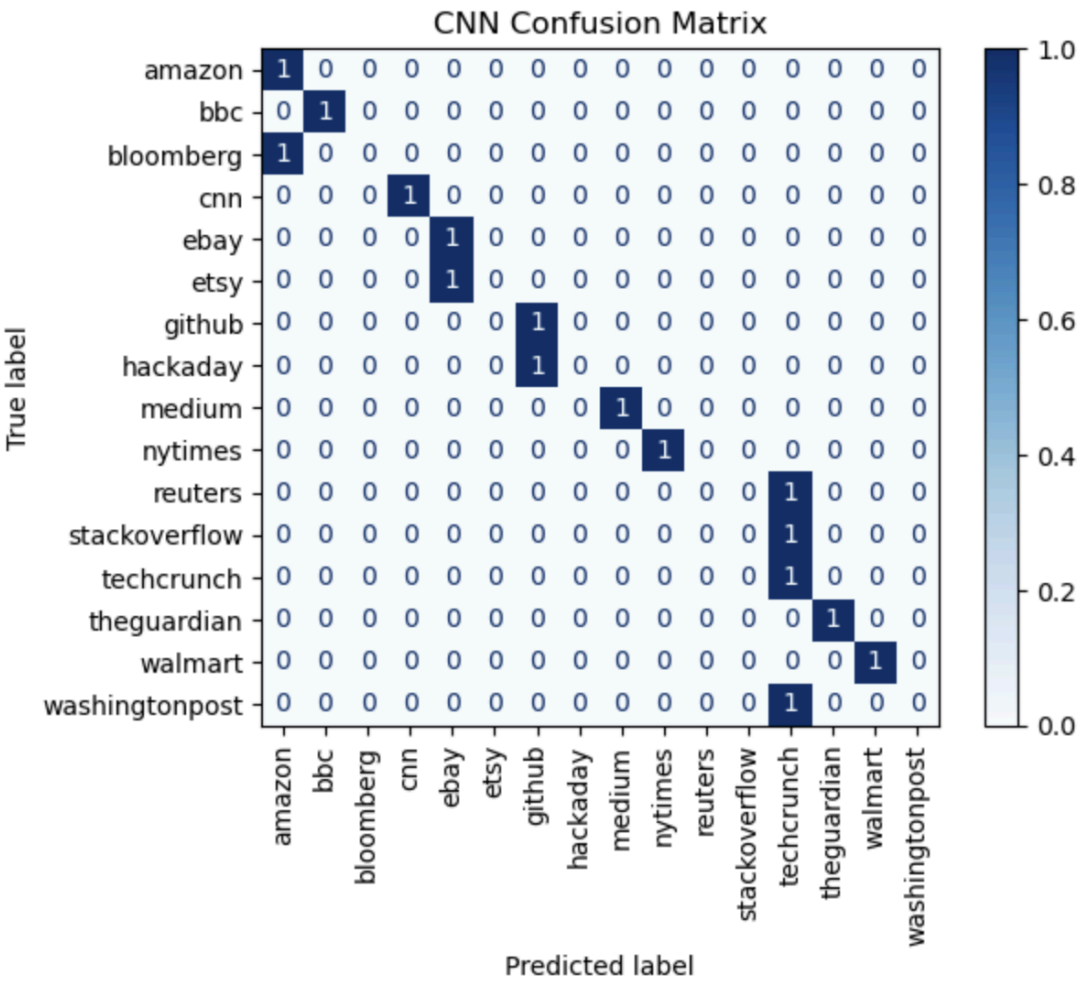

True label
Predicted label
amazon
bbc
bloomberg
cnn
ebay
etsy
github
hackaday
medium
nytimes
reuters
stackoverflow
techcrunch
theguardian
walmart
washingtonpost
0
1
2
3
4
5

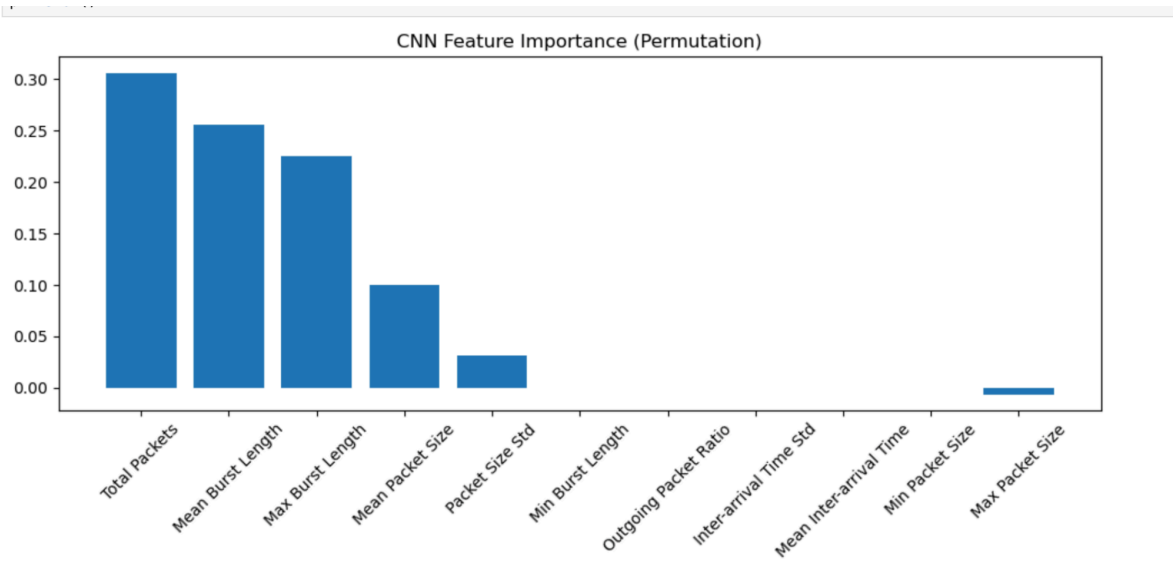

SVM Feature Importance (Permutation)
min_burst
max_burst
mean_burst
outgoing_ratio
std_inter_time
mean_inter_time
min_packet_size
max_packet_size
std_packet_size
mean_packet_size
total_packets
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.16

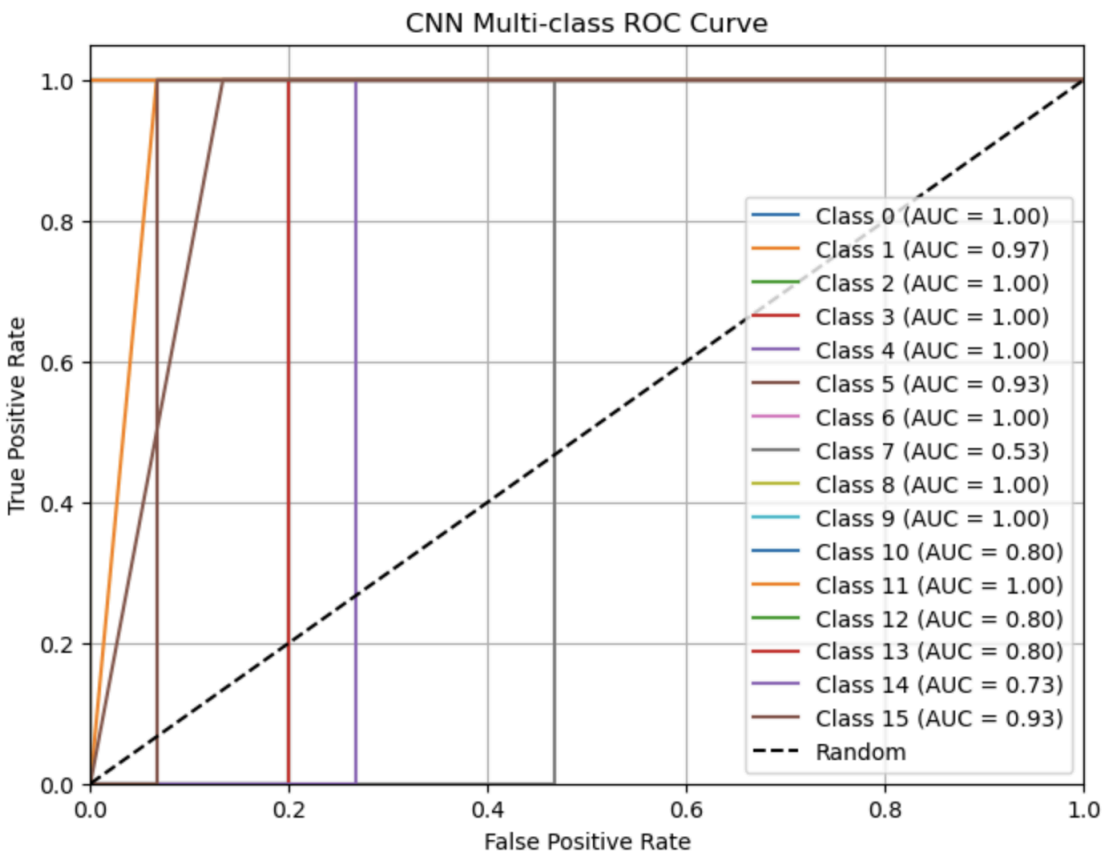

ROC Curve — SVM
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Macro-average ROC (AUC = 0.91)
Random classifier

CNN:


train_acc
val_acc
Accuracy
Epoch
0.0
0.1
0.2
0.3
0.4
0.5
0
5
10
15
20
25
30


CNN Confusion Matrix
True label
Predicted label
amazon
bbc
bloomberg
cnn
ebay
etsy
github
hackaday
medium
nytimes
reuters
stackoverflow
techcrunch
theguardian
walmart
washingtonpost
1.0
0.8
0.6
0.4
0.2
0.0


CNN Feature Importance (Permutation)
0.30
0.25
0.20
0.15
0.10
0.05
0.00
Total Packets
Mean Burst Length
Max Burst Length
Mean Packet Size
Packet Size Std
Min Burst Length
Outgoing Packet Ratio
Inter-arrival Time Std
Mean Inter-arrival Time
Min Packet Size
Max Packet Size



NPZ 1 file

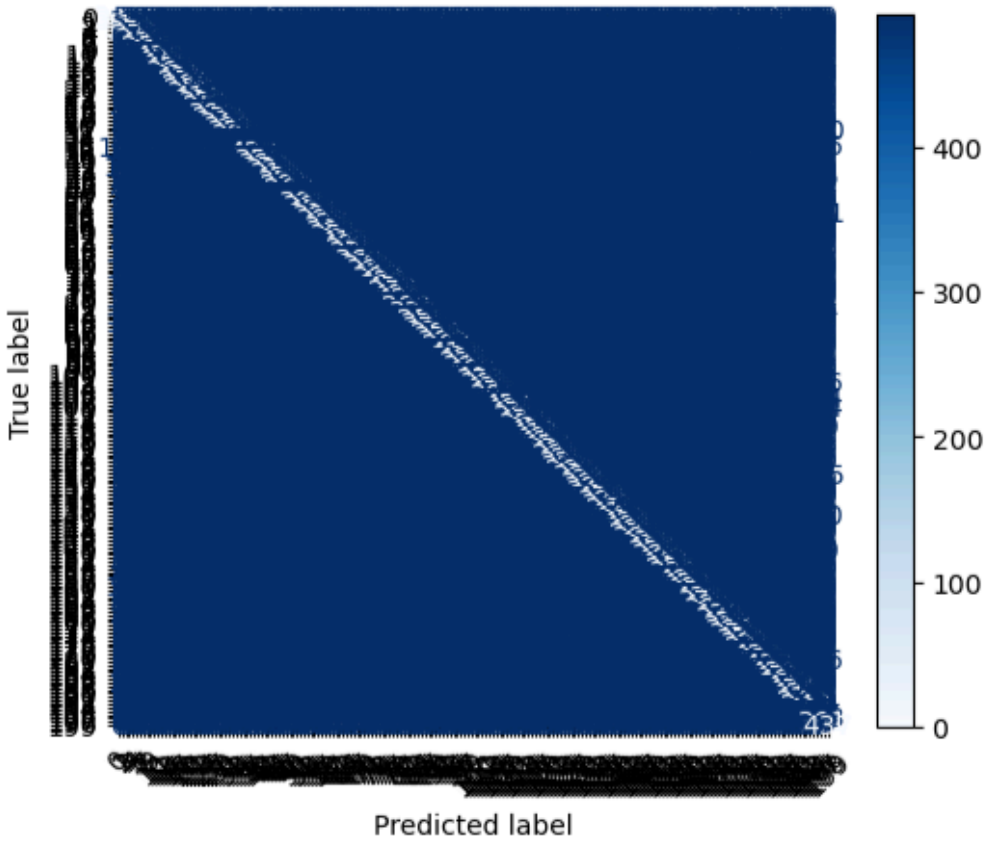
	RF	**SVM**	**CNN**
Accuracy	0.71114	0.7668	0.6921
Precision	0.7302783823819717	0.7974	0.702
Recall	0.71114	0.7668	0.6921
F1-score	0.7017996126249845	0.7716	0.6819
Confusion matrix comment			
ROC-curve			

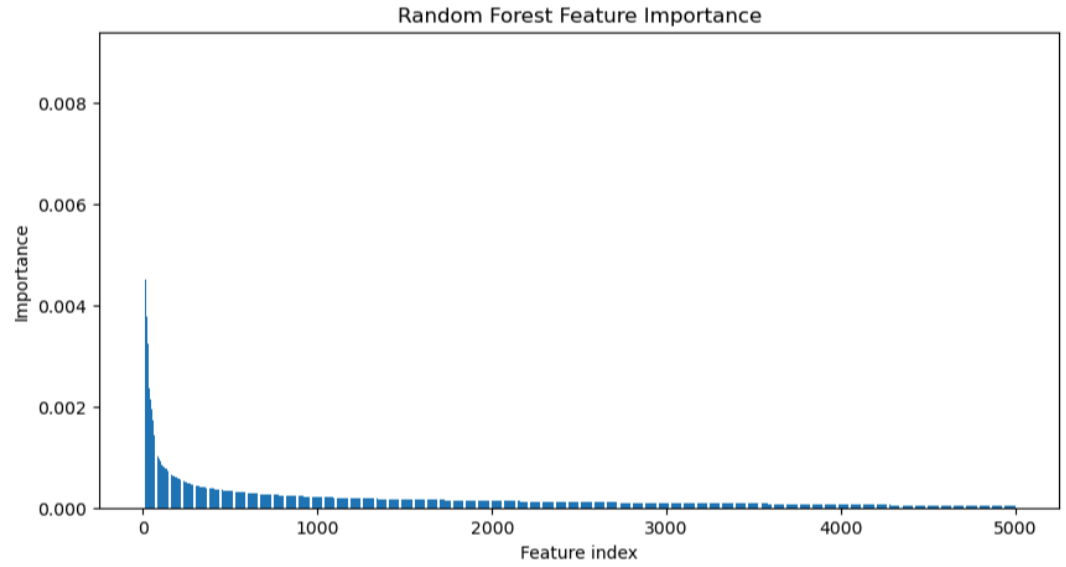
Random Forest

Комментарий:

- Модель Random Forest демонстрирует умеренно высокий уровень точности (Accuracy) и сбалансированные показатели Precision и Recall, что указывает на хорошее распознавание большинства классов.
- Значение Precision выше Recall, что говорит о том, что среди предсказанных положительных примеров есть небольшое количество ложных срабатываний, но большинство предсказаний верны.
- F1-score (среднее гармоническое Precision и Recall) составляет около 70%, что отражает общий компромисс между точностью и полнотой.
- При учете большого числа классов (200 сессий) и высокой размерности признаков (5000) данный результат показывает, что Random Forest способен справляться с задачей многоклассовой классификации, хотя модель всё еще может путать похожие сессии.
- Для улучшения показателей можно рассмотреть: уменьшение размерности признаков (feature selection / PCA), увеличение числа деревьев, настройку гиперпараметров или использование ансамблей моделей.

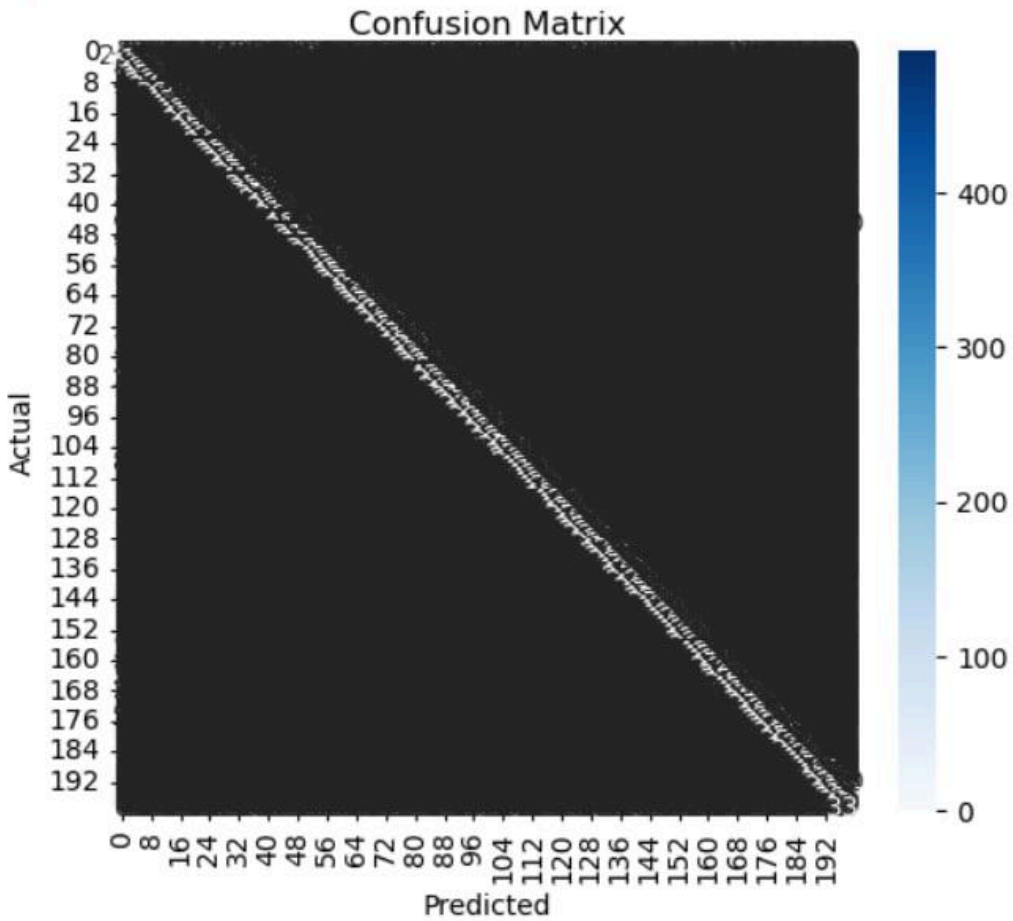
Confusion matrix





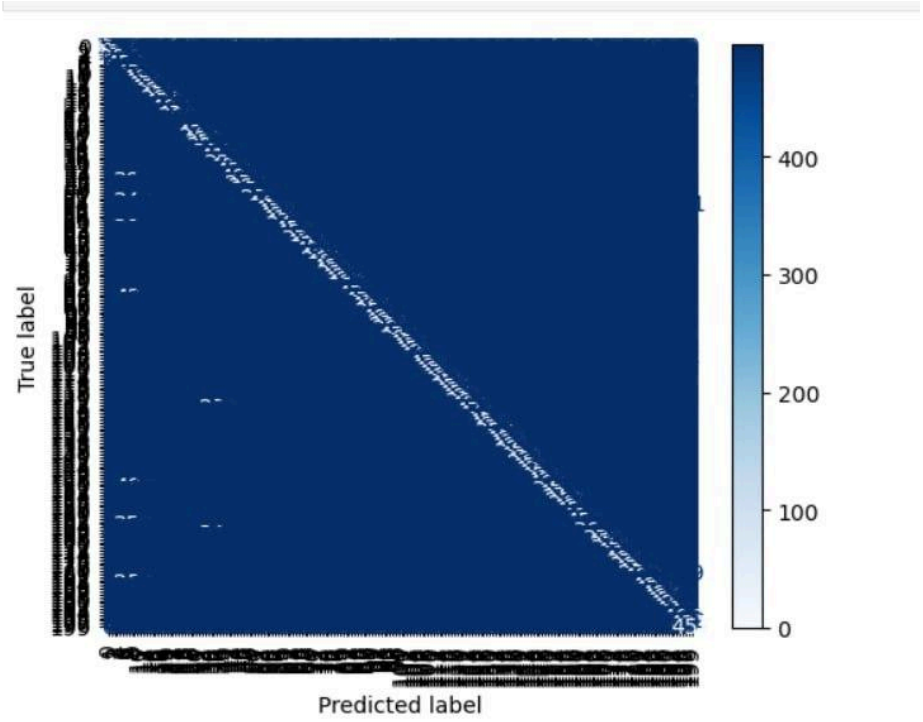
SGD (Linear SVM)

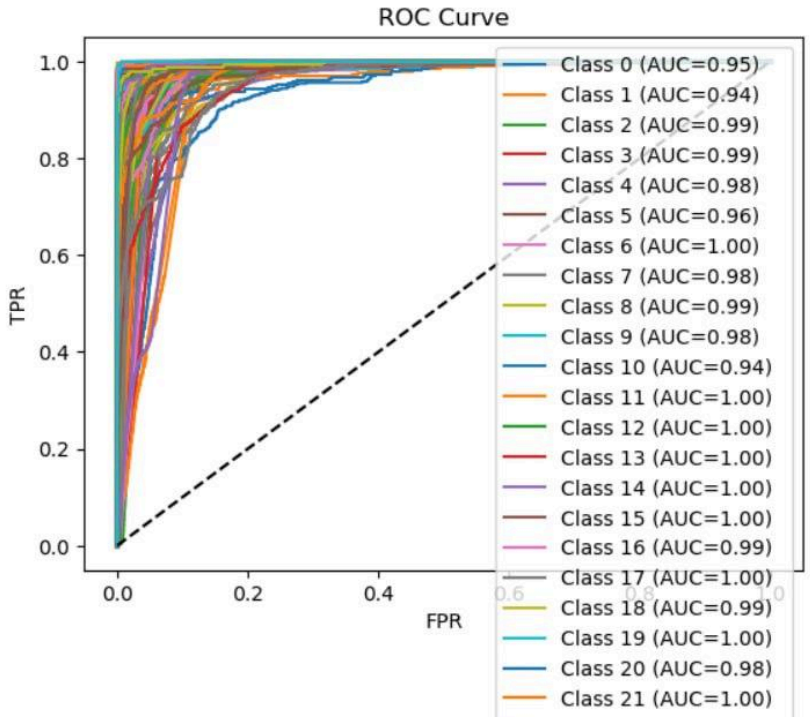
Accuracy: 0.76682
Precision: 0.7973834401341368
Recall: 0.76682
F1-score: 0.7716414686094348



CNN for npz

CNN Accuracy: 0.692080020904541
CNN Precision: 0.7020470816045149
CNN Recall: 0.69208
CNN F1-score: 0.6818992803772744




ROC Curve
TPR
FPR
0.0
0.2
0.4
0.6
0.8
1.0
Class 0 (AUC=0.95)
Class 1 (AUC=0.94)
Class 2 (AUC=0.99)
Class 3 (AUC=0.99)
Class 4 (AUC=0.98)
Class 5 (AUC=0.96)
Class 6 (AUC=1.00)
Class 7 (AUC=0.98)
Class 8 (AUC=0.99)
Class 9 (AUC=0.98)
Class 10 (AUC=0.94)
Class 11 (AUC=1.00)
Class 12 (AUC=1.00)
Class 13 (AUC=1.00)
Class 14 (AUC=1.00)
Class 15 (AUC=1.00)
Class 16 (AUC=0.99)
Class 17 (AUC=1.00)
Class 18 (AUC=0.99)
Class 19 (AUC=1.00)
Class 20 (AUC=0.98)
Class 21 (AUC=1.00)